

Pre-A Q3LOCK Local Renyi, Doublet, and Prospective Freeze Checkpoint

TECT verification-first repository
claim C6-SPACETIME-SIGNATURE

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1 Purpose and scope

This single gate-level synthesis records EXP-000806 and EXP-000807, together with the R-168 validator-hardening correction EXP-000808. It advances reusable result R-167 to additive version v1.9 and registers R-168 v1.0. Their stable identifiers are, respectively,

1. [PA-CP1-ST8-Q3LOCK-SECOND-WEIGHTED-ENERGY-MOMENT-AND-COMMON-ALPHA-CAUCHY-GATE-SPLIT](#);
2. [PA-ROUND1-PROSPECTIVE-HOLDOUT-FREEZE-PROTOCOL-AND-CURRENT-TREE-READINESS-AUDIT](#).

The task is T-054, the claim context is C6-SPACETIME-SIGNATURE, both results are T0, and both have `claim_bearing: false`. No claim-card tier changes in this checkpoint.

R-167 v1.9 proves three narrow statements: exact pure-bond coordinate-tail invariance with a state-weighted cutoff identity; for each fixed finite region S , a reduction from a local coordinate-marginal measured-Renyi likelihood bound to the corresponding two-orientation Gaussian weighted tail; and, for fixed positive coupling ratio and sufficiently small semiclassical parameter, the isolated Q3 onsite doublet theorem together with the exact rank-two transverse-field Ising compression and explicit low/high residual data. It also proves that one volume-independent *global* all-bond sandwiched-Renyi target is overstrong in a conditional low-doublet product fixture.

R-168 v1.0 proves a fail-closed schema-shape and provenance-order protocol for a future prospective Round-1 holdout. It audits the registered snapshot at commit 99157442831c0e44d425b5d5f8cd78856c57da53. That snapshot has zero official freeze records, zero admitted microscopic survivors, and no admitted M1, M2, or M5 map/prediction pair. The initially observed local `freeze/*` tag count was zero, but that is a non-load-bearing live observation, not an audited-commit fact, blocker, PASS condition, remote receipt, or value required to remain zero. R-168 creates no target, official freeze record, frozen prediction, score, public freeze commit, or annotated freeze tag.

The two mathematical parents and the prospective parent remain open. In particular, this checkpoint proves no onsite-interspersed local history likelihood estimate, no all-exhaustion common alpha, no rank-two unbounded band block diagonalization, no two-phase QPS transfer for the exact oscillator lattice, no actual broken-sector temporal mass or GNS gap, and no cryptographic custodian-signature or independent remote-ref verification. It proves no regulator removal, continuum, same-Hamiltonian physical-empty comparison, physical vacuum selection, C6, CP1, physical Sector A, or Pre-A closure.

2 Provenance and exact gate ledger

The formal and analytic authorities for R-167 v1.9 are

1. [strategy/pre-a-cp1-st8-q3lock-local-measured-renyi-semiclassical-doublet-route-split-manifest.json](#);
2. [strategy/pre-a-cp1-st8-q3lock-local-measured-renyi-semiclassical-doublet-route-split-certificate-260811.md](#).

Its direct exploration parents are EXP-000789 and EXP-000805; its current decision is EXP-000806. The three subgates closed in v1.9 are exactly

1. [PA-CP1-ST8-Q3LOCK-PURE-BOND-COORDINATE-TAIL-INVARIANCE-AND-STATE-WEIGHTED-CUTOFF-IDENTITY](#);
2. [PA-CP1-ST8-Q3LOCK-LOCAL-MEASURED-RENYI-TO-HISTORY-TAIL-REDUCTION](#);
3. [PA-CP1-ST8-Q3LOCK-SEMICLASSICAL-ONSITE-DOUBLET-AND-EXACT-LOW-BAND-TFIM-COMPRESSION](#).

The two new scoped negative results are

1. [NG-2026-08-11-PRE-A-ST8-Q3LOCK-GLOBAL-ALL-BOND-RENYI-VOLUME-UNIFORMITY](#);
2. [NG-2026-08-11-PRE-A-ST8-Q3LOCK-RANK-ONE-UNBOUNDED-BLOCK-DIAGONALIZATION-DIRECT-BROKEN-DOUBLET-IMPORT](#).

The formal and analytic authorities for R-168 v1.0 are

1. [strategy/pre-a-round1-prospective-holdout-freeze-protocol-manifest.json](#);
2. [strategy/pre-a-round1-prospective-holdout-freeze-protocol-certificate-260811.md](#).

Its direct parent is EXP-000791; its initial decision is EXP-000807, and the append-only validator-hardening correction is EXP-000808. The four protocol/audit subgates closed in v1.0 are exactly

1. [PA-ROUND1-COMMON-ESTIMAND-AND-CANDIDATE-MAP-SCHEMA](#);
2. [PA-ROUND1-PROSPECTIVE-FREEZE-PROVENANCE-PROTOCOL](#);
3. [PA-ROUND1-TARGET-INDEPENDENCE-AND-ANTI-LEAKAGE-SCHEMA-VALIDATOR](#);
4. [PA-ROUND1-CURRENT-CANDIDATE-MAP-ADMISSION-EMPTY-SET-AUDIT](#).

Its current-checkpoint negative is [NG-2026-08-11-PRE-A-ROUND1-CURRENT-TREE-PROSPECTIVE-HOLDOUT-NONEXISTENCE](#). This is not a no-go for a future independently committed target or a future admitted microscopic map.

The live open gates relevant to this synthesis are

1. [PA-CP1-ST8-Q3LOCK-LOCAL-STRICT-ALL-EXHAUSTION-TWO-ORIENTATION-HISTORY-COMMON-ALPHA](#);
2. [PA-CP1-ST8-Q3LOCK-BROKEN-SECTOR-GNS-GAP-COERCIVITY](#);
3. [PA-CP1-ST8-Q3LOCK-INFINITE-DIMENSIONAL-RANK-TWO-BAND-BLOCK-DIAGONALIZATION-AND-TWO-PHASE-QPS](#);
4. [PA-ROUND1-EVIDENCE-ROLE-AND-MINIMUM-MANIFEST-FREEZE](#);

5. [PA-ROUND1-PER-PARAMETER-COMMON-INPUT-LEDGER](#);
6. [PA-ROUND1-INDEPENDENT-CUSTODIAN-OPAQUE-TARGET-COMMITMENT](#);
7. [PA-ROUND1-ADMISSIBLE-MICROSCOPIC-CANDIDATE-MAP-AND-FROZEN-PREDICTION](#);
8. [PA-ROUND1-CRYPTOGRAPHIC-CUSTODIAN-SIGNATURE-AND-REMOTE-FREEZE-VERIFICATION](#).

The earlier fixed-Trotter compatibility, phasewise GNS-gap/OS-mass equivalence, and conditional doublet-Ising reference-gap reductions remain available at their registered scopes. They are not restated as new closures here.

3 Exact pure-bond coordinate-tail theorem

Write the full cross-bond multiplication operator and its unitary layer as

$$V_{\times} = -c \sum_{\langle xy \rangle} q_x \cdot q_y, \quad B_{\delta} = \exp(-i\delta V_{\times}/\hbar). \quad (3.1)$$

Every bounded coordinate multiplier $F(q_S)$ commutes with B_{δ} . Hence, in both history orientations,

$$B_{\delta}^* F(q_S) B_{\delta} = F(q_S). \quad (3.2)$$

This includes the local tail projection

$$E_{S,L} = 1_{\{X_S > L\}}, \quad X_S = \max_{x \in S} |q_x|, \quad (3.3)$$

and, by strong commutation of affiliated multiplication operators, weighted coordinate multipliers on their natural finite-moment forms.

Let $V_{\times,L}$ be a coordinate cutoff and put

$$W_L = V_{\times} - V_{\times,L}, \quad B_{\delta,L} = \exp(-i\delta V_{\times,L}/\hbar). \quad (3.4)$$

All three multiplication operators commute. For every finite-volume type-I normal state σ satisfying $\sigma(W_L^2) < \infty$, functional calculus gives the exact two-orientation Hilbert–Schmidt identity

$$\begin{aligned} \|(B_{\delta} - B_{\delta,L})\sigma^{1/2}\|_2^2 &= \|\sigma^{1/2}(B_{\delta} - B_{\delta,L})\|_2^2 \\ &= \sigma\left(4 \sin^2 \frac{\delta W_L}{2\hbar}\right) \\ &\leq \frac{\delta^2}{\hbar^2} \sigma(W_L^2). \end{aligned} \quad (3.5)$$

Therefore a state-weighted all-bond telescope pays no artificial $\exp(CL^2)$ operator-norm multiplier at a pure bond layer. Equation (3.5) does not estimate $\sigma(W_L^2)$ in the actual intermediate states. The missing input is precisely the two-orientation W_L^2 tail after onsite and bond layers have been interspersed.

4 Why the global all-bond Renyi target is overstrong

Use the conditional low-doublet product reference

$$\rho_1 = \begin{pmatrix} 4/5 & 0 \\ 0 & 1/5 \end{pmatrix}, \quad \rho_2 = \rho_1 \otimes \rho_1, \quad (4.1)$$

with compressed coordinate $q_e = m\sigma_x$. One complete eight-component spatial bond acts as

$$U_\theta = \exp(i\theta \sigma_x \otimes \sigma_x), \quad \theta = \frac{8\delta cm^2}{\hbar} = \frac{\delta J}{\hbar}, \quad J = 8cm^2. \quad (4.2)$$

The angle $\delta cm^2/\hbar$ is only one internal-coordinate channel, not the full bond. Direct matrix multiplication gives

$$\tilde{Q}_2(U_\theta \rho_2 U_\theta^* || \rho_2) = \frac{(4 + 9 \sin^2 \theta)^2}{16}. \quad (4.3)$$

At $\theta = \pi/4$ the exact value is

$$\frac{289}{64}; \quad (4.4)$$

on three disjoint bonds tensor multiplicativity gives

$$\frac{24137569}{262144}. \quad (4.5)$$

For every fixed $\theta \notin \pi\mathbb{Z}$, (4.3) therefore grows exponentially in the number of disjoint bonds. Nevertheless every spectral function of the *compressed* coordinate $m\sigma_x$ commutes with the kick, so its probability is unchanged in this fixture. This is not an identity for $P1_{\{q>L\}}P$ in the full oscillator.

The counterfixture rejects only a volume-independent global sandwiched-Renyi bound for complete all-bond layers in this conditional product reference. It does not refute a full interacting Q3 Gibbs local measured-Renyi theorem, a restricted-tail domination theorem, or the existence of common dynamics.

5 Local measured-Renyi sufficient condition

Fix finite S , choose $\alpha > 1$, and define

$$\vartheta = \frac{\alpha - 1}{\alpha}. \quad (5.1)$$

Let $\nu_{0,S}$ be the reference law of X_S , and let $\nu_{P,S}^+$ and $\nu_{P,S}^-$ be its laws after a partial history and its adjoint. The minimal direct target is

$$\sup_{u \geq L_0} \frac{\nu_{P,S}^\pm(X_S > u)}{\nu_{0,S}(X_S > u)^\vartheta} \leq C_{\alpha,T,S}, \quad (5.2)$$

uniformly in volume, source, mesh, partial history, and orientation. A stronger checkable condition confined to the commutative coordinate algebra is

$$\int \left(\frac{d\nu_{P,S}^\pm}{d\nu_{0,S}} \right)^\alpha d\nu_{0,S} \leq Q_{\alpha,T,S}. \quad (5.3)$$

Condition (5.3) is weaker than a global quantum sandwiched-Renyi bound. If

$$\nu_{0,S}(X_S > u) \leq M_a |S| e^{-au^2}, \quad (5.4)$$

Holder's inequality yields the two-orientation probability estimate

$$\nu_{P,S}^+(X_S > L) + \nu_{P,S}^-(X_S > L) \leq 2Q_{\alpha,T,S}^{1/\alpha} (M_a |S|)^\vartheta e^{-\vartheta a L^2}. \quad (5.5)$$

Putting $b = \vartheta a$, fourth-moment layer cake gives

$$2Q_{\alpha,T,S}^{1/\alpha}(M_a|S|)^{\vartheta} e^{-bL^2} \left(L^4 + \frac{2L^2}{b} + \frac{2}{b^2} \right). \quad (5.6)$$

This closes only the local-tail implication for each fixed finite S . It does not by itself supply the full growing weighted-tail corridor. That corridor still requires either a bound uniform over translated two-site shapes, or an explicit growth envelope for Q_{α,T,S_R} that remains absorbable when the history region S_R , range R , and cutoff L grow. No onsite-interspersed Q3LOCK estimate of the form (5.2) or (5.3) is proved here.

6 Q3 onsite semiclassical theorem

Set

$$R = -r > 0, \quad \mu = \frac{\lambda}{g} > 0, \quad v = \sqrt{R/g}, \quad E_* = \frac{R^2}{g}, \quad h = \frac{\hbar g}{\sqrt{\chi} R^{3/2}}. \quad (6.1)$$

After $q = vx$ and the registered additive shift, the one-site operator is

$$h_{\text{site}} = E_* K_h, \quad K_h = -\frac{h^2}{2} \Delta + W_\mu, \quad (6.2)$$

where

$$W_\mu(x) = \frac{1}{4} \sum_{e=1}^8 (x_e^2 - 1)^2 + \frac{\mu}{4} \sum_{\{e,f\} \in E(Q_3)} (x_e - x_f)^2 (x_e^2 + x_f^2). \quad (6.3)$$

For fixed $\mu > 0$, W_μ is coercive and has exactly the two global minima $\pm \mathbf{1}$. At either minimum,

$$\text{Hess } W_\mu = 2I + \mu L_{Q_3}, \quad (6.4)$$

with spectrum

$$2, \quad (2 + 2\mu)^{\times 3}, \quad (2 + 4\mu)^{\times 3}, \quad 2 + 6\mu. \quad (6.5)$$

Both wells are nondegenerate, and the exact dimensionless Agmon action is

$$S_0 = \frac{16\sqrt{2}}{3}. \quad (6.6)$$

The load-bearing primary sources are Simon's nondegenerate-minimum and degenerate-cluster results, including Theorem 1.1, Theorem 4.1, Lemma 4.2, Theorem 5.1, Theorem 5.3, and Corollary 5.4 in https://www.numdam.org/item/AIHPA_1983__38_3_295_0.pdf; the same-parity correction in https://www.numdam.org/item/AIHPA_1984__40_2_224_0/; the multiple-well analysis at <https://doi.org/10.1080/03605308408820335>; Simon's two-well logarithmic splitting theorem, Theorem 1.5, at <https://annals.math.princeton.edu/1984/120-1/p04>; and the symmetry analysis at https://www.numdam.org/item/AIHPA_1985__42_2_127_0/. The Q3-specific certificate verifies the hypotheses and normalization before using those sources.

For every fixed $\mu > 0$, there is a non-explicit $h_0(\mu) > 0$ such that $0 < h < h_0(\mu)$ gives simple first and second eigenvalues. Their normalized vectors ϕ_0, ϕ_1 are, respectively, even and odd under global sign reversal and are $\text{Aut}(Q_3)$ singlets. If

$$\delta_1 = \epsilon_1 - \epsilon_0, \quad \Gamma = \epsilon_2 - \epsilon_0, \quad (6.7)$$

then

$$\lim_{h \downarrow 0} h \log \frac{\delta_1}{E_*} = -S_0, \quad (6.8)$$

and, for every $\eta > 0$,

$$\delta_1 \leq E_* C_{\mu, \eta} e^{-(S_0 - \eta)/h}. \quad (6.9)$$

The harmonic expansion gives

$$\frac{\Gamma}{E_*} = \sqrt{2}h + O(h^{3/2}), \quad (6.10)$$

and

$$\frac{\epsilon_0}{E_*} = h e_{\text{well}} + O(h^{3/2}), \quad e_{\text{well}} = \frac{1}{2} \left[\sqrt{2} + 3\sqrt{2+2\mu} + 3\sqrt{2+4\mu} + \sqrt{2+6\mu} \right]. \quad (6.11)$$

After fixing the phase of ϕ_1 , cube symmetry and weighted harmonic localization give, for every coordinate e ,

$$\begin{aligned} m &= \langle \phi_0, q_e \phi_1 \rangle = v[1 + O(h)], \\ a_j &= \langle \phi_j, q_e^2 \phi_j \rangle = v^2[1 + O(h)], \\ b_j &= \langle \phi_j, q_e^4 \phi_j \rangle = v^4[1 + O(h)], \\ a_1 - a_0 &= O(v^2 h). \end{aligned} \quad (6.12)$$

The last line is deliberately only the safe algebraic localization consequence. No exponential rate for the even-observable difference is imported without an additional weighted Agmon-overlap theorem. All constants are existential; this theorem does not certify a numerical h_0 for the repository's finite $r = -9$ diagnostic.

7 Exact low-band compression and residual inputs

Let

$$P = |\phi_0\rangle\langle\phi_0| + |\phi_1\rangle\langle\phi_1|, \quad P_1 = |\phi_1\rangle\langle\phi_1|, \quad s = |\phi_0\rangle\langle\phi_1| + \text{h.c.} \quad (7.1)$$

and put $d_2 = a_1 - a_0$. Parity and cube symmetry give

$$P q_e P = m s, \quad P q_e^2 P = \text{diag}(a_0, a_1). \quad (7.2)$$

For

$$B_{xy} = \frac{c}{2} \sum_{e=1}^8 (q_{xe} - q_{ye})^2, \quad (7.3)$$

the exact compression to $P_\Lambda = P^{\otimes \Lambda}$, after one scalar shift, is

$$P_\Lambda H P_\Lambda = J \sum_{\langle xy \rangle} (1 - s_x s_y) + \sum_x [\delta_1 + 4c \deg(x) d_2] P_{1,x}, \quad qquad J = 8cm^2. \quad (7.4)$$

On a periodic cubic lattice of degree $z = 6$,

$$\delta_{\text{eff}} = \delta_1 + 24c(a_1 - a_0). \quad (7.5)$$

Open finite boxes instead have the exact site-dependent boundary field in (7.4); a uniform coefficient $24c$ is not asserted there.

Let $Q = 1 - P$ and define

$$a^2 = \max_{j=0,1} (a_j - m^2) = \|Qq_e P\|^2, \quad b^2 = \max_{j=0,1} (b_j - a_j^2) = \|Qq_e^2 P\|^2. \quad (7.6)$$

One bond obeys the exact reduction

$$\|(1 - P_x P_y) B_{xy} P_x P_y\| \leq 8c[b + 2ma + a^2]. \quad (7.7)$$

With

$$k = h_{\text{site}} - \epsilon_0 - \delta_1 P_1 \geq \Gamma Q, \quad \mathcal{R}_e = q_e - ms, \quad (7.8)$$

every $t > 0$ satisfies

$$\|\mathcal{R}_e \psi\|^2 \leq (1+t)a^2 \|P\psi\|^2 + (1+t^{-1})A_Q \langle Q\psi, kQ\psi \rangle, \quad (7.9)$$

where

$$A_Q = \frac{v^2}{\Gamma} + \frac{2\sqrt{\epsilon_0 + \Gamma}}{\Gamma\sqrt{g}}. \quad (7.10)$$

These formulas reduce, but do not solve, the high-mode control needed for a volume-uniform band theorem.

7.1 A nonempty asymptotic model corridor

Take

$$g = \lambda = \chi = \hbar = 1, \quad r = -N^4, \quad c = N^{-4}. \quad (7.11)$$

Then

$$v = N^2, \quad E_* = N^8, \quad h = N^{-6}, \quad J \longrightarrow 8, \quad \Gamma \sim \sqrt{2}N^2, \quad (7.12)$$

and the certified asymptotic bookkeeping is

$$\begin{aligned} a &= O(N^{-1}), & b &= O(N), \\ \|(1 - P_x P_y) B_{xy} P_x P_y\| &= O(N^{-3}), \\ cA_Q &= O(N^{-2}), & cm\sqrt{A_Q} &= O(N^{-1}), \\ \delta_{\text{eff}} &= O(N^{-6}) + O\left(N^8 e^{-(S_0 - \eta)N^6}\right), \\ A_0 &\sim \frac{2}{9}N^4. \end{aligned} \quad (7.13)$$

Thus no algebraic conflict is forced between the registered infrared lower condition, an order-one Ising scale, and vanishing low/high ratios. This is an asymptotic family, not a finite- N enclosure, a block-diagonalization theorem, or a QPS theorem.

8 Boundary of the unbounded block import

Del Vecchio, Frohlich, and Pizzo prove a volume-uniform Lie–Schwinger theorem for short-range unbounded form-bounded interactions:

S. Del Vecchio, J. Frohlich, and A. Pizzo, *Block-diagonalization of infinite-volume lattice Hamiltonians with unbounded interactions*, JFA 284 (2023) 109734, <https://arxiv.org/pdf/2108.13907>.

Its stated theorem uses a rank-one onsite vacuum, tensor products of rank-one local reference projectors, and concludes a unique ground state with a uniform gap. The introductory degenerate-extension remark is not a proved theorem for a rank-two onsite band whose global low space has dimension $2^{|\Lambda|}$.

Choosing ϕ_0 alone does not repair the mismatch: its onsite gap δ_1 is exponentially small while the desired Ising scale J remains order one in (7.12). The successor must be a volume-uniform rank-two band Lie–Schwinger/Feshbach theorem that subtracts (7.4) and leaves a sufficiently small finite-dimensional interaction for a two-phase QPS theorem, or an equivalent spectral-cutoff removal theorem preserving both phases and a common positive gap.

The registered negative rejects only the direct rank-one import. It is not a no-go for a future rank-two theorem and not a counterexample to an exact Q3LOCK broken-sector gap.

9 Prospective target, map, and freeze objects

R-168 separates the candidate-neutral target contract from every candidate-specific microscopic map. The first object is

$$\mathcal{T} = (\text{estimand, units, raw estimator, disclosure rule, independence group}). \quad (9.1)$$

For each microscopic candidate M_i , the second object is

$$\mathcal{M}_i : (\text{microscopic parameters, state, reference, limits}) \mapsto \mathcal{T}. \quad (9.2)$$

An admitted map must state its theorem, domain, state, reference, units map, order of limits, nuisance inputs, proof references, executable script, normalized source hash, and admission status. Maps marked `ABSENT`, `CONDITIONAL_POSTHOC`, or `NOT_ADMITTED` are ineligible. The unique M0 effective baseline is mandatory but cannot win as a microscopic theory. At least one actual microscopic contestant must have both an admitted map and a nonempty frozen physical prediction.

The schema is `tect/pre-a-round1-prospective-holdout-freeze/1.0`, with exact root fields

```
schema, freeze_id, prediction_id, freeze_version, round_id, parent_gate,  
status, claim_bearing, fixture_only, contestant_snapshot, evidence_snapshot,  
target_contract, observable_contract, prediction_contract,  
robustness_contract, provenance, scoring, no_overclaim
```

A valid real freeze has lifecycle state `FROZEN_UNSCORED`, is claim-nonbearing, and is not a synthetic fixture. The exact target fields are

```
target_id, custodian, protocol_or_accession, estimand_id, units,  
independence_group, blind, predictor_access_before_freeze,  
target_value_present, commitment, disclosure
```

The exact commitment and disclosure fields are

algorithm, commitment_hex, secret_key_custody, payload_schema,
canonical_serialization, domain_separation, custodian_signature,
public_key_fingerprint, issued_at_utc

status, not_before_utc, actual_at_utc

The exact prediction fields are

candidate_id, predicted_relation, physical_output, theory_uncertainty,
acceptance_rule, baseline_prediction, allowed_inputs, forbidden_knobs

and every allowed-input row has exactly

id, class, source, source_id, used_for

The robustness contract has exactly `volume`, `boundary`, `regulator`, `coefficients`, and `implementation` as nonempty textual frozen envelopes. It is not represented as a separately hash-pinned robustness program. Contestant manifests, evidence snapshots, admitted map scripts, the raw estimator, and the scorer are normalized-SHA-256 pinned.

Before semantic comparison, every declared leaf has its exact Boolean, positive-integer, nonempty-string, digest/OID, UTC-time, null, or enum type, and every declared container has its exact object/list type. Root, contestant, evidence, target, commitment, disclosure, observable, estimand, raw-estimator, candidate-map, prediction, allowed-input, robustness, provenance, and scoring objects enforce their declared structural keys. The root and target identity strings are nonempty. The target `estimand_id` and `units` equal the common estimand. Candidate maps are unique by `candidate_id`; the singular frozen prediction has a nonempty relation and names a score-eligible contestant backed by an admitted map.

The syntactic firewall rejects declared target values, registered target-bearing aliases, and undeclared payload fields at the exact-controlled structural locations. It does not prove that arbitrary allowed free text is free of steganographically encoded target information. That semantic independence remains an external custodian-attestation and review obligation.

10 Temporal and cryptographic provenance boundary

The required order is

$$t_{\text{custodian}} \leq t_{\text{public freeze}} < t_{\text{disclosure}} \leq t_{\text{score}}. \quad (10.1)$$

Let T be the target payload with schema `tect/pre-a-round1-holdout-target-payload/1.0`, let $\text{JCS}(T)$ be its RFC 8785 canonicalization, and let $D = \text{TCT-PRE-A-ROUND1-HOLDOUT-TARGET-v1}$. The required commitment is

$$C = \text{HMAC-SHA256}(K_{\text{custodian}}, D \parallel 0x00 \parallel \text{JCS}(T)). \quad (10.2)$$

Here `0x00` is one zero octet and the exact custody field is `EXTERNAL_CUSTODIAN`; an author-held key or plain unkeyed hash is not accepted. The custodian signs the canonical commitment metadata, including payload schema, canonicalization, domain separator, digest, and issue time.

A future public anchor identifies the remote repository, exact remote commit object and observation time, annotated tag `freeze/<PRED-ID>/v<N>`, distinct annotated-tag object, remote ref, custodian signature, and authenticated receipt. The remote URL must parse as HTTPS with a nonempty hostname.

An unpushed local commit, filesystem modification time, or self-declared `issued_on` value is not prospective temporal evidence. The present validator checks schema shape and local hash pins only. It does not verify a signature against a public key and does not fetch or validate the remote commit, tag

object, or ref. Every purported real freeze is therefore rejected with `EXTERNAL_VERIFICATION_REQUIRED`; changing `fixture_only` from true to false cannot promote a fixture.

Every local hash pin uses a normalized relative POSIX repository path. Drive paths, absolute paths, backslashes, dot segments, parent traversal, and paths resolving outside the repository are rejected. This local containment check does not prove that the blob occurs in the remote freeze commit; that binding belongs to the independent remote verifier.

Discovery and fitting evidence IDs must be disjoint from the holdout independence group. Each input is classified as one of

`INSERTED`, `FITTED`, `MATCHED`, `CALIBRATION`, `DERIVED`, `PREDICTED`,
`VISIBLE_VALIDATION`, `NOT_AVAILABLE`.

Each allowed input has nonempty `source` and `source_id`; its `source_id` must be absent from both discovery and forbidden-fit IDs. Exact row keys reject undeclared source-ID aliases. These checks make declared prohibited reuse detectable. They do not make an input scientifically valid by schema alone or prove semantic independence hidden inside free text.

11 Hostile fixtures and exact current-tree audit

The fail-closed executable uses synthetic fixture data and rejects all 28 registered hostile classes:

1. an undeclared root payload: `ROOT_FIELDS_EXTRA`;
2. empty root identities: `ROOT_VALUES_INVALID`;
3. empty target identities: `TARGET_CONTRACT_INVALID`;
4. target/common-estimand ID or units mismatch: `TARGET_ESTIMAND_MISMATCH`;
5. a direct target value or interval: `TARGET_LEAKAGE`;
6. a hidden `sealed_payload`: `TARGET_LEAKAGE`;
7. a `holdout_value` alias: `TARGET_LEAKAGE`;
8. invalid disclosure ordering: `TEMPORAL_ORDER_INVALID`;
9. a mutated candidate, evidence, map, estimator, or scorer hash: `HASH_FAILURE`;
10. a parent-traversing or repository-escaping hash path: `HASH_FAILURE`;
11. independence-group overlap: `INDEPENDENCE_OVERLAP`;
12. a missing unique M0 baseline: `BASELINE_MISSING`;
13. duplicate candidate maps: `MAP_CANDIDATE_DUPLICATE`;
14. an eligible candidate without an admitted map: `ELIGIBLE_MAP_MISSING`;
15. a prediction not bound to an eligible admitted-map candidate: `PREDICTION_CANDIDATE_UNBOUND`;
16. a missing allowed-input `source_id`: `INPUT_FIELDS_INVALID`;
17. discovery-source reuse: `DISCOVERY_REUSE`;
18. an undeclared `discovery_source_id` alias: `INPUT_FIELDS_INVALID`;

19. a malformed nested commitment: `COMMITMENT_INVALID`;
20. list-valued signatures or integer-valued commitment digests: `COMMITMENT_INVALID`;
21. integer-valued commit or tag object IDs: `REMOTE_ANCHOR_INVALID`;
22. an HTTPS URL with no parsed hostname: `REMOTE_ANCHOR_INVALID`;
23. a list-valued input class or null `used_for`: `INPUT_LEDGER_INVALID`;
24. a non-string common-estimand definition: `ESTIMAND_INVALID`;
25. a non-string contestant role: `CONTESTANTS_INVALID`;
26. a non-object common estimand: `ESTIMAND_INVALID`;
27. a non-object raw estimator: `ESTIMAND_INVALID`;
28. malformed or absent anchor metadata: `REMOTE_ANCHOR_INVALID`.

Malformed nested values return structured rejection reports rather than escaping as type, attribute, set-membership, or hashing exceptions. The valid fixture remains synthetic and `fixture_only: true`; no hostile suite result promotes it to a real freeze.

At audited commit `99157442831c0e44d425b5d5f8cd78856c57da53`, the executable reconstructs the registered inputs from commit blobs. It finds

$$N_{\text{records}} = 0, \quad N_{\text{admitted microscopic survivors}} = 0. \quad (11.1)$$

The stable audit has no tag count. Separately, $N_{\text{tags}}^{\text{local,initial}} = 0$ was a non-load-bearing live observation at initial execution. It is not encoded by the audited commit, is not a blocker, is not a remote query or cryptographic receipt, and is not required to remain zero. A future legitimate tag may change it without invalidating the historical audit. M1 has no admitted microscopic map and no frozen physical prediction. M2 has an empty registered `physical_predictions` list and a false holdout flag; its retrospective conditional mapping cannot become a prospective test. M5 has no admitted map and no holdout prediction.

The exact blocker set is

```
NO_MACHINE_FREEZE_RECORD
NO_ADMITTED_MICROSCOPIC_SURVIVOR
M1_MAP_AND_PREDICTION_ABSENT
M2_PHYSICAL_PREDICTION_AND_HOLDOUT_ABSENT
M5_MAP_AND_HOLDOUT_ABSENT
PER_PARAMETER_COMMON_INPUT_LEDGER_INCOMPLETE
PROSPECTIVE_PREDICTION_NOT_FROZEN
```

Thus `actual_freeze_ready=false` from exactly seven stable blockers at the audited checkpoint. The live tag observation does not enter this Boolean. The negative result is snapshot-specific and does not claim that a future map, prediction, commitment, or freeze is impossible.

12 Exact completion conditions

Before target disclosure, the prospective parent can advance only after all of the following exist:

1. an independent custodian's externally keyed, signed canonical HMAC-SHA-256 target commitment;
2. a candidate-neutral estimand and hash-pinned raw estimator;

3. at least one admitted candidate-specific microscopic map;
4. a nonempty physical prediction with a complete common-input ledger;
5. a frozen baseline, uncertainty, and acceptance rule;
6. a public remote commit and separately authorized annotated freeze tag;
7. cryptographic signature validation and independent remote object/ref verification with an authenticated receipt;
8. an independent implementation and declared robustness envelope; and
9. later execution of the unchanged scorer after documented disclosure.

The target is external information. Repository proof work cannot manufacture it or retroactively repair prospective time ordering.

The next dynamics action is a volume-, source-, mesh-, partial-history-, and orientation-uniform local measured likelihood or restricted-tail theorem for onsite-interspersed Q3LOCK histories. The next gap action is a volume-uniform rank-two unbounded-band elimination or equivalent cutoff removal theorem, followed by a two-phase QPS transfer and beta-infinity phase identification. The next prospective action is an independently held target commitment together with at least one admitted microscopic map and frozen nonempty prediction, completed before disclosure.

13 Exact dependency boundary for Sector A and Pre-A

The dynamics parent is

OPEN, with fixed-Trotter compatibility, pure-bond coordinate-tail invariance, and local measured-Renyi sufficiency isolated. The actual onsite-interspersed two-orientation history bound and all-exhaustion common alpha remain open.

The spectral parent is

OPEN, with the fixed- μ small- h onsite doublet, exact low-band TFIM compression, and centered residual inputs isolated. Rank-two unbounded-band elimination, two-phase transfer, phase selection, and the actual broken-sector GNS gap remain open.

The prospective parent is

OPEN, with the schema, provenance order, anti-leakage shape validator, and current-tree empty-admission audit closed. No external target commitment, admitted microscopic prediction, real freeze, remote anchor verification, or score exists.

The active A5-SECTOR-A-SYNTHESIS remains a T6 conditional composition under its registered hypotheses. Nothing here promotes it to a physical or full-Class-II Sector-A theorem. Physical Pre-A additionally requires the registered target-algebra ground-state selection, actual sector coercivity, enlarged-counterterm continuum control, same-Hamiltonian physical-empty comparison, and genuinely prospective validation. Those obligations remain open.

14 Devil's-advocate review

1. **UPHELD as an open obligation:** exact pure-bond invariance does not control intermediate states after onsite layers. The two-orientation W_L^2 history tail is still missing.
2. **DISMISSED:** the bond telescope has not hidden an $\exp(CL^2)$ factor. Equation (3.5) is an exact state-weighted identity; its unresolved quantity is displayed explicitly.
3. **UPHELD as a scope boundary:** the global Renyi no-go is a conditional two-level product fixture, not a full interacting Q3 Gibbs counterexample and not a dynamics nonexistence theorem.
4. **DISMISSED:** the full eight-component bond factor is not missing. Equation (4.2) uses $\theta = 8\delta cm^2/\hbar = \delta J/\hbar$.
5. **UPHELD as an open obligation:** the semiclassical threshold $h_0(\mu)$ is existential. The finite $r = -9$ diagnostic is not certified by the imported theorem.
6. **UPHELD as a rate boundary:** only $a_1 - a_0 = O(v^2h)$ is used. No unproved exponential weighted-overlap rate is inserted into δ_{eff} .
7. **DISMISSED:** the coefficient $24c$ is not silently extended to open boundaries. It is stated only for the periodic degree-six cubic lattice; equation (7.4) retains the site-dependent field.
8. **UPHELD as an import mismatch:** the published rank-one theorem does not supply a rank-two two-phase theorem. The negative rejects the direct import, not the possible successor.
9. **DISMISSED:** the freeze validator is not represented as a signature or remote-ref verifier. Purported real freezes fail closed with `EXTERNAL_VERIFICATION_REQUIRED`.
10. **UPHELD as a semantic boundary:** exact keys and types reject declared target fields, aliases, and undeclared payloads, but cannot prove that allowed free text contains no steganographically encoded target. That requires external custodian attestation and review.
11. **DISMISSED:** the initially observed zero live-tag count is not used as a stable blocker or PASS condition. A future legitimate tag can change that informational value without invalidating the historical audit.
12. **DISMISSED:** seven narrow subgate closures do not close C6, CP1, physical Sector A, or Pre-A. The open parents are repeated in Sections 2, 12, and 13.

15 Reproduction and evidence contract

The R-167 v1.9 primary, non-importing independent, and integrated engines are

1. `codes/foundations/pre_a_cp1_st8_q3lock_local_measured_renyi_semiclassical_doublet_route_split.py`;
2. `codes/foundations/pre_a_cp1_st8_q3lock_local_measured_renyi_semiclassical_doublet_route_split_independent.py`;
3. `codes/foundations/pre_a_cp1_st8_q3lock_local_measured_renyi_semiclassical_doublet_route_split_verify.py`.

The R-168 v1.0 primary, non-importing independent, and integrated engines are

1. `codes/foundations/pre_a_round1_prospective_holdout_freeze_protocol.py`;
2. `codes/foundations/pre_a_round1_prospective_holdout_freeze_protocol_independent.py`;
3. `codes/foundations/pre_a_round1_prospective_holdout_freeze_protocol_verify.py`.

From the repository root, run

```
Set-Location E:\Dev\TECT
$py = 'E:\Dev\TECT.venv\Scripts\python.exe'
$r167 = (
    'codes\foundations\pre_a_cp1_st8_q3lock_' +
    'local_measured_renyi_semiclassical_doublet_route_split'
)
& $py -X utf8 ($r167 + '.py')
& $py -X utf8 ($r167 + '_independent.py')
& $py -X utf8 ($r167 + '_verify.py')
$r168 = (
    'codes\foundations\pre_a_round1_' +
    'prospective_holdout_freeze_protocol'
)
& $py -X utf8 ($r168 + '.py')
& $py -X utf8 ($r168 + '_independent.py')
& $py -X utf8 ($r168 + '_verify.py')
```

The expected R-167 proof-layer outputs are primary PASS 96/96, independent PASS 81/81, and integrated PASS 181/181. The expected R-168 outputs are primary PASS 125/125, independent PASS 153/153, and integrated PASS 218/218. The integrated totals are the staged targets for this source; they become final checkpoint evidence only when each integrated engine confirms the completed source, stored-result freshness, formal authorities, import firewall, deterministic reruns, and cross-engine constants. A later source-check failure invalidates the corresponding integrated total rather than weakening the check.

This v0.8 file is the only synthesis source drafted for this logical checkpoint. At source-drafting time no sibling PDF is asserted. A PDF may be issued only after the integrated source checks pass, followed by one gate-level compile and all-page render review under the repository PDF economy rule. No per-lemma or intermediate PDF belongs to these results.

16 Result footer

Result ID:	PA-CP1-ST8-Q3LOCK-SECOND-WEIGHTED-ENERGY-MOMENT-AND-COMMON-ALPHA-CAUCHY-GATE-SPLIT; PA-ROUND1-PROSPECTIVE-HOLDOUT-FREEZE-PROTOCOL-AND-CURRENT-TREE-READINESS-AUDIT.
Reusable results:	R-167 v1.9; R-168 v1.0.
Explorations:	EXP-000806; EXP-000807; EXP-000808 correction.
Precise statement:	Exact pure-bond state-weighted cutoff identity; fixed-finite-S measured-Renyi tail implication; fixed-mu, small-h Q3 onsite doublet; exact low-band TFIM compression and residuals; prospective-freeze exact-key/type protocol; current-tree audit.
Closed subgates:	Three R-167 mathematical reductions and four R-168 schema/audit gates, exactly as listed in Section 2.
Open parents:	All-exhaustion two-orientation history common alpha; broken-sector GNS-gap coercivity; rank-two unbounded band/QPS transfer; actual prospective freeze.
Negative results:	No volume-uniform global all-bond Renyi target in the conditional doublet product fixture; no direct rank-one block-theorem import; no actual holdout at

the audited current-tree checkpoint.

Scope: T0 exact route reductions, theorem import after Q3 hypothesis checks, protocol schema, and audit; claim_bearing false.

Dependencies: EXP-000789, EXP-000791, EXP-000805; R-167 v1.0-v1.8; fixed-mu small-h semiclassics; audited commit 99157442831c0e44d425b5d5f8cd78856c57da53; external custodian and remote verifier remain open.

Evidence grade: ANALYTIC + EXACT + primary + non-importing independent + integrated verification, conditional on the final integrated source/freshness checks.

Reproduction command: Run the six commands in Section 15 in the listed order, with each integrated verifier last.

Expected output: R-167: PASS 96/96, PASS 81/81, PASS 181/181.
R-168: PASS 125/125, PASS 153/153, PASS 218/218.

Falsification gate: Any sign, factor-eight bond angle, Hilbert-Schmidt orientation, Renyi exponent, layer-cake coefficient, Q3 Hessian multiplicity, action, semiclassical rate, TFIM field, residual coefficient, snapshot count, hostile-fixture verdict, provenance, freshness, hash, source, count, or scope mismatch.

Tier before / after: C6 T1 / C6 T1; no claim or tier change.

No-overclaim statement: No onsite-interspersed likelihood bound, split-limit completion, all-exhaustion common alpha, rank-two block diagonalization, two-phase QPS transfer, beta-infinity selection, actual temporal mass/GNS gap, cryptographic or remote freeze verification, semantic free-text target-independence proof, target, prediction, tag, score, physical-empty sign, regulator removal, continuum, C6, CP1, physical Sector A, or Pre-A closure.

Next required action: Prove the local history-tail theorem and the rank-two band/QPS theorem; independently obtain the external commitment, admitted microscopic map, frozen nonempty prediction, and authenticated remote verification before disclosure.